

# Muhammad Anas

**Data and Analytics Engineer | Python, SQL, ETL Pipelines, Tableau, Power BI**

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## SUMMARY

AI Consultant at Convergent Business Technologies since 2024, building AI features, integrations and multi tenant backends into client systems in banking, retail, marketing analytics and higher education. Built a ZATCA Phase 2 e invoicing gateway for a Middle East bank (all 6 ZATCA APIs, ECDSA signing, hash chaining), the NestJS backend of a marketing mix modeling platform with Microsoft Entra ID and row level security, and an AI recruitment portal that screens resumes with Gemini. Earlier, an AI assisted retail reporting pipeline for UAE and KSA cut the monthly cycle from 7 days to about 2 hours.

## SKILLS

**Programming:** Python, TypeScript, SQL, FastAPI, NestJS, Next.js, React

**AI and Automation:** LLM features and workflow automation, generative AI and LLMs, Gemini API including vision, RAG and prompt pipelines, AI agents, n8n, API integration

**Backend and Data:** PostgreSQL with row level security, multi tenant design, TypeORM, Supabase, ETL data pipelines, relational SQL database design

**Auth, Security and Compliance:** Microsoft Entra ID, OAuth, email OTP second factor, role based access control, ECDSA signing, ZATCA e invoicing integration

**Cloud and Tools:** Docker, Git, Vercel, Render, Cloudflare R2, QStash, Postman, Jest, Odoo, Zoho, Notion

**BI:** Tableau, Power BI, Microsoft Excel

**Practices:** test strategy and QA, technical documentation, C4 architecture models, requirement gathering

## EXPERIENCE

### Convergent Business Technologies, Pakistan

2024 to present

*AI Consultant*

- Build AI features, integrations and backends for client products in banking, retail, marketing analytics and higher education
- Implemented AI assisted workflow automation using LLMs and n8n, reducing processing time by 60 to 75 percent
- Developed automated ETL pipelines using Python, SQL and Excel, and designed centralized SQL databases to remove version conflicts
- Built dashboards and reports using Tableau and Power BI
- Wrote architecture documents, API references and test plans, and collaborated with business, audit and IT teams to ensure system traceability

## PROJECTS

### PepsiCo Retail Analytics, UAE and KSA

2024

*Stack: Python, Tableau, Gemini, Docker, Excel*

- Built an end to end monthly reporting pipeline for UAE and KSA retail data, harmonizing SKU level sales, quantity, volume, manufacturer and competitor detail across 13 retailers into one role based platform
- Automated SKU matching using Python and AI assisted logic as one step in the pipeline, reducing matching time from 80 minutes to 40 seconds per 100 SKUs, accuracy up from 70 to 95 percent
- Automated the full monthly refresh, database push, Tableau dashboard rebuild and Excel report generation, with all numbers validated automatically before delivery
- Cut the full monthly reporting cycle from 7 days to about 2 hours, and data validation turnaround from 3 days to 1

### ZTBL Credit Risk Model Automation

2025

*Stack: Python, SQL, Docker, Excel*

- Automated 20+ Excel based PD, LGD, EAD and ECL models using Python and SQL, reducing manual effort by about 75 percent
- Designed and deployed a relational SQL database to centralize inputs, eliminating version conflicts

### Marketing Mix Modeling Platform, Backend

2026

*Stack: NestJS, TypeScript, PostgreSQL, TypeORM, Microsoft Entra ID, Cloudflare R2, Render*

- Built the NestJS backend of a multi tenant marketing mix modeling platform in a four person team, owning authentication, data, uploads and model orchestration APIs
- Implemented Microsoft Entra ID token authentication, email OTP as a second factor, and invite only projects with three tier permissions
- Designed multi tenant PostgreSQL with row level security and separate app and admin database roles, managed through TypeORM migrations

- Integrated two modeling engines, Google Meridian and PyMC Marketing, through train, status and results endpoints, plus data quality checks, channel health, exposure metrics and collinearity checks with a ridge fallback
- Wrote the API reference, architecture document and Postman collection; 156 commits, all 26 backend test cases passing

### **ZATCA Phase 2 E Invoicing Gateway and Middleware, Banking and ERP**

2026

*Stack: TypeScript, Next.js, node forge, xmlbuilder2, Supabase, QStash, Vercel, Odoo, Zoho*

- Built a ZATCA Phase 2 e invoicing gateway for a Middle East bank, implementing all 6 ZATCA APIs: compliance CSID, compliance invoice checks, production CSID, clearance, reporting and renewal
- Implemented UBL 2.1 invoice XML, ECDSA secp256k1 signing, SHA 256 invoice hash chaining and QR code generation, and fixed double hashing and previous invoice hash encoding bugs
- Designed a multi bank isolated architecture, then extended it into a multi tenant middleware for Odoo and Zoho with a headless OpenAPI and pluggable adapters
- Wrote the C4 architecture model and developer and user manuals; launched and deployed for use

### **CBT Recruitment Workflow Automation**

2024

*Stack: Python, Notion, n8n, LLMs*

- Consolidated multiple recruitment workflows into a single Notion based platform
- Automated candidate assessment steps using workflow automation and LLMs, reducing coordination time by 60 percent
- Piloted AI agents for repetitive tasks, reducing average response time by 2 days

### **AI Recruitment Portal, Full Stack**

2025 to 2026

*Stack: Next.js, TypeScript, Tailwind CSS, Supabase, Gemini API, Vercel*

- Built and own a production recruitment portal for a graduate academy program, from application to assessment
- Screens resumes with Gemini 2.0 Flash against admin defined criteria, using vision for image only PDFs
- Implemented role based access for four roles, Microsoft corporate sign in, assessment slot booking, a reapply cooldown and audit logging

## **EDUCATION**

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Iqra University, Islamabad. Bachelor of Science, Computer Science, 2019 to 2023.